Understanding Language with Machines

Binary Text Classification Using Logistic Regression

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Motivation

Common Applications









Why Understanding Language is Hard

Humans understanding meaning → machines only interpret numbers

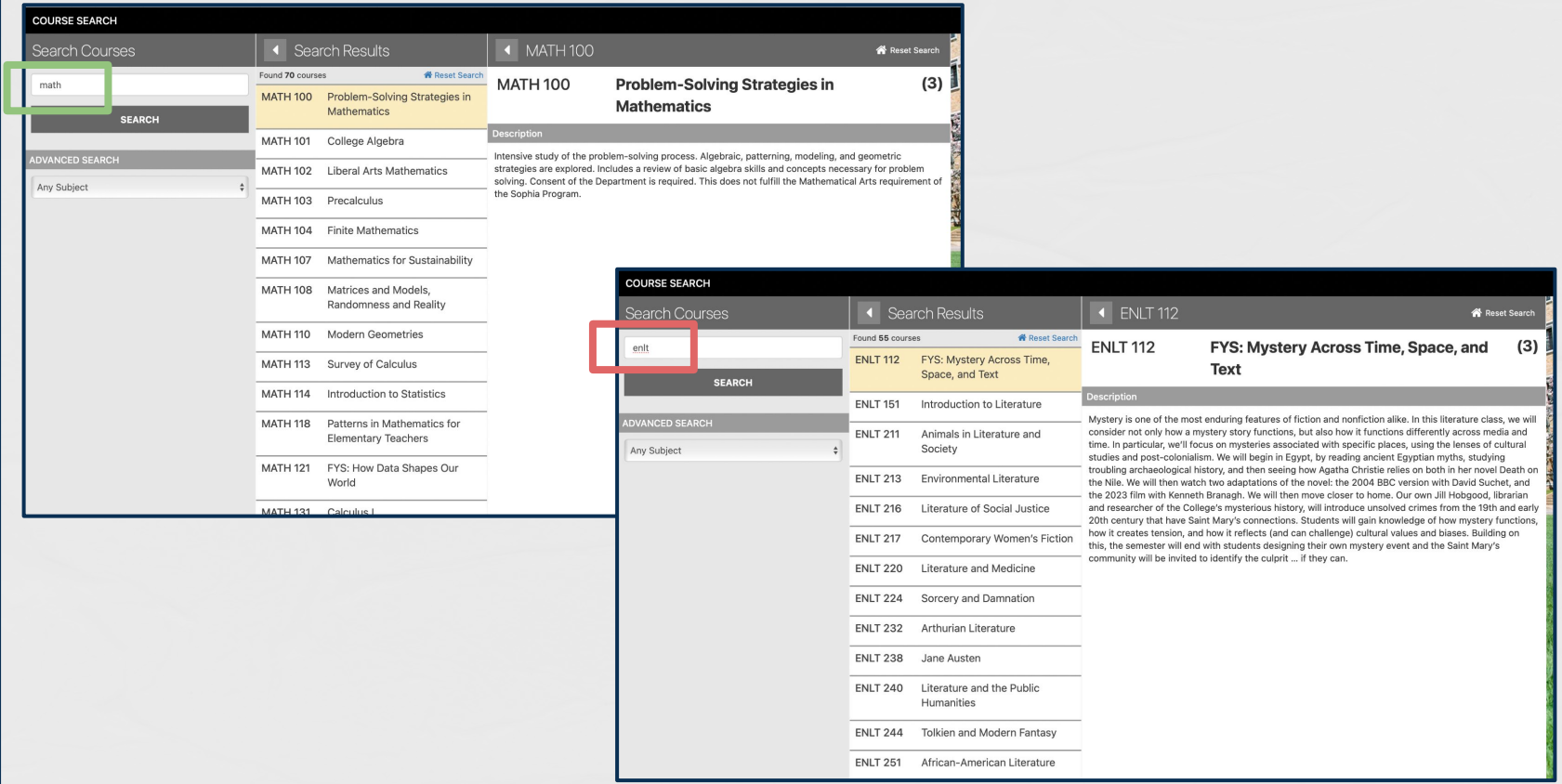
There are plenty of ways to represent text → we chose **TF-IDF**

Goal of the Project

Understand the mathematical principles that underlie more complex language models

1. Understand logistic regression
2. Applying the TF-IDF method
3. Implementing Binary Classifier
4. Analyze model performance and coefficients

The Dataset


COURSE SEARCH
Search Courses
math
SEARCH
ADVANCED SEARCH
Any Subject
Search Results
Found 70 courses
Reset Search
MATH 100 Problem-Solving Strategies in Mathematics
MATH 101 College Algebra
MATH 102 Liberal Arts Mathematics
MATH 103 Precalculus
MATH 104 Finite Mathematics
MATH 107 Mathematics for Sustainability
MATH 108 Matrices and Models, Randomness and Reality
MATH 110 Modern Geometries
MATH 113 Survey of Calculus
MATH 114 Introduction to Statistics
MATH 118 Patterns in Mathematics for Elementary Teachers
MATH 121 FYS: How Data Shapes Our World
MATH 131 Calculus I
MATH 100
Reset Search
MATH 100 Problem-Solving Strategies in Mathematics (3)
Description
Intensive study of the problem-solving process. Algebraic, patterning, modeling, and geometric strategies are explored. Includes a review of basic algebra skills and concepts necessary for problem solving. Consent of the Department is required. This does not fulfill the Mathematical Arts requirement of the Sophia Program.
COURSE SEARCH
Search Courses
enlt
SEARCH
ADVANCED SEARCH
Any Subject
Search Results
Found 55 courses
Reset Search
ENLT 112 FYS: Mystery Across Time, Space, and Text
ENLT 151 Introduction to Literature
ENLT 211 Animals in Literature and Society
ENLT 213 Environmental Literature
ENLT 216 Literature of Social Justice
ENLT 217 Contemporary Women's Fiction
ENLT 220 Literature and Medicine
ENLT 224 Sorcery and Damnation
ENLT 232 Arthurian Literature
ENLT 238 Jane Austen
ENLT 240 Literature and the Public Humanities
ENLT 244 Tolkien and Modern Fantasy
ENLT 251 African-American Literature
ENLT 112
Reset Search
ENLT 112 FYS: Mystery Across Time, Space, and Text (3)
Description
Mystery is one of the most enduring features of fiction and nonfiction alike. In this literature class, we will consider not only how a mystery story functions, but also how it functions differently across media and time. In particular, we'll focus on mysteries associated with specific places, using the lenses of cultural studies and post-colonialism. We will begin in Egypt, by reading ancient Egyptian myths, studying troubling archaeological history, and then seeing how Agatha Christie relies on both in her novel Death on the Nile. We will then watch two adaptations of the novel: the 2004 BBC version with David Suchet, and the 2023 film with Kenneth Branagh. We will then move closer to home. Our own Jill Hobgood, librarian and researcher of the College's mysterious history, will introduce unsolved crimes from the 19th and early 20th century that have Saint Mary's connections. Students will gain knowledge of how mystery functions, how it creates tension, and how it reflects (and can challenge) cultural values and biases. Building on this, the semester will end with students designing their own mystery event and the Saint Mary's community will be invited to identify the culprit ... if they can.

The Dataset

label	course_id	title	description
ENLT	ENLT 106W	Language and Li	A range of courses taught in tandem with "W" courses in other disciplines. Students may earn three literature hours and fulfill the w
ENLT	ENLT 112	FYS: Mystery Ac	Mystery is one of the most enduring features of fiction and nonfiction alike. In this literature class, we will consider not only how a m
ENLT	ENLT 151	Intoduction to Lit	Introduction to Literature (3/4) This course introduces students to the skills of reading and writing that enable us to appreciate, und
ENLT	ENLT 211	Animals in Literat	This interdisciplinary course explores how literature shapes and reflects human responses to nonhuman animals, using theoretical
ENLT	ENLT 213	Environmental Li	This course studies writings that engage their readers with the natural environment. We will learn the ways of reading that these te
ENLT	ENLT 216	Literature of Soci	What role do literary works play in advocating for a more just society? Can literature provide a platform for social exploration? An o
ENLT	ENLT 217	Contemporary W	Readings of contemporary women writers since the 1960s.
ENLT	ENLT 220	Literature and M	What can literature tell us about experiences of health, illness, and health care? How do literary works represent and even interven
ENLT	ENLT 224	Sorcery and Dan	From Homer and Dante to Anthony Burgess and Anne Rice, this course examines one of the oldest and most fascinating of literary
ENLT	ENLT 232	Arthurian Literatu	An exploration of the myth of Arthur from its medieval roots to the present day. While this course will focus on literary depictions of
MATH	MATH 100	Problem-Solving	Intensive study of the problem-solving process. Algebraic, patterning, modeling, and geometric strategies are explored. Includes a
MATH	MATH 101	College Algebra	This course will cover topics in algebra that are needed for future courses. Topics will include basic algebraic concepts, linear equa
MATH	MATH 102	Liberal Arts Math	Mathematical modeling through the use of graph theory. Topics include graphs, directed graphs, trees, matchings and network flow
MATH	MATH 103	Precalculus	This course studies polynomial, rational, exponential, logarithmic, and trigonometric functions from the symbolic, numeric, and grap
MATH	MATH 104	Finite Mathemati	Set theory, counting techniques, probability, random variables, expected value, variance, standard deviation, and linear programmi
MATH	MATH 107	Mathematics for	This course develops and applies mathematical concepts and tools to quantitatively explore sustainability issues. Topics such as in
MATH	MATH 108	Matrices and Mo	Matrices, systems of equations, determinants, eigenvalues, linear transformations, complex numbers, orthogonality. Emphasis on a
MATH	MATH 110	Modern Geometr	Finite geometries. Transformational geometry with an introduction to fractals. Euclidean geometry, including classical constructions
MATH	MATH 113	Survey of Calcul	A one semester introduction to differential and integral calculus designed primarily for liberal arts students and those in the professi
MATH	MATH 114	Introduction to St	Introduction to basic sampling and experimental design. Basics of probability, random variables, and probability distributions. Samp

Core Concepts

Preprocessing

Raw text → numerical features

Standardize text and reduce noise

MATH 103: PRE-CALCULUS

This course studies polynomial, rational, exponential, logarithmic, and trigonometric functions from the symbolic, numeric, and graphical perspectives. The emphasis on these concepts will provide solid preparation for a college-level calculus course. This does not fulfill the Mathematical Arts requirement of the Sophia Program. Prerequisite: MATH 101 or recommendation of Math Placement Advisor.



```
[course, studies, polynomial, rational, exponential, logarithmic,
    trigonometric, functions, ..., college-level, calculus]
```

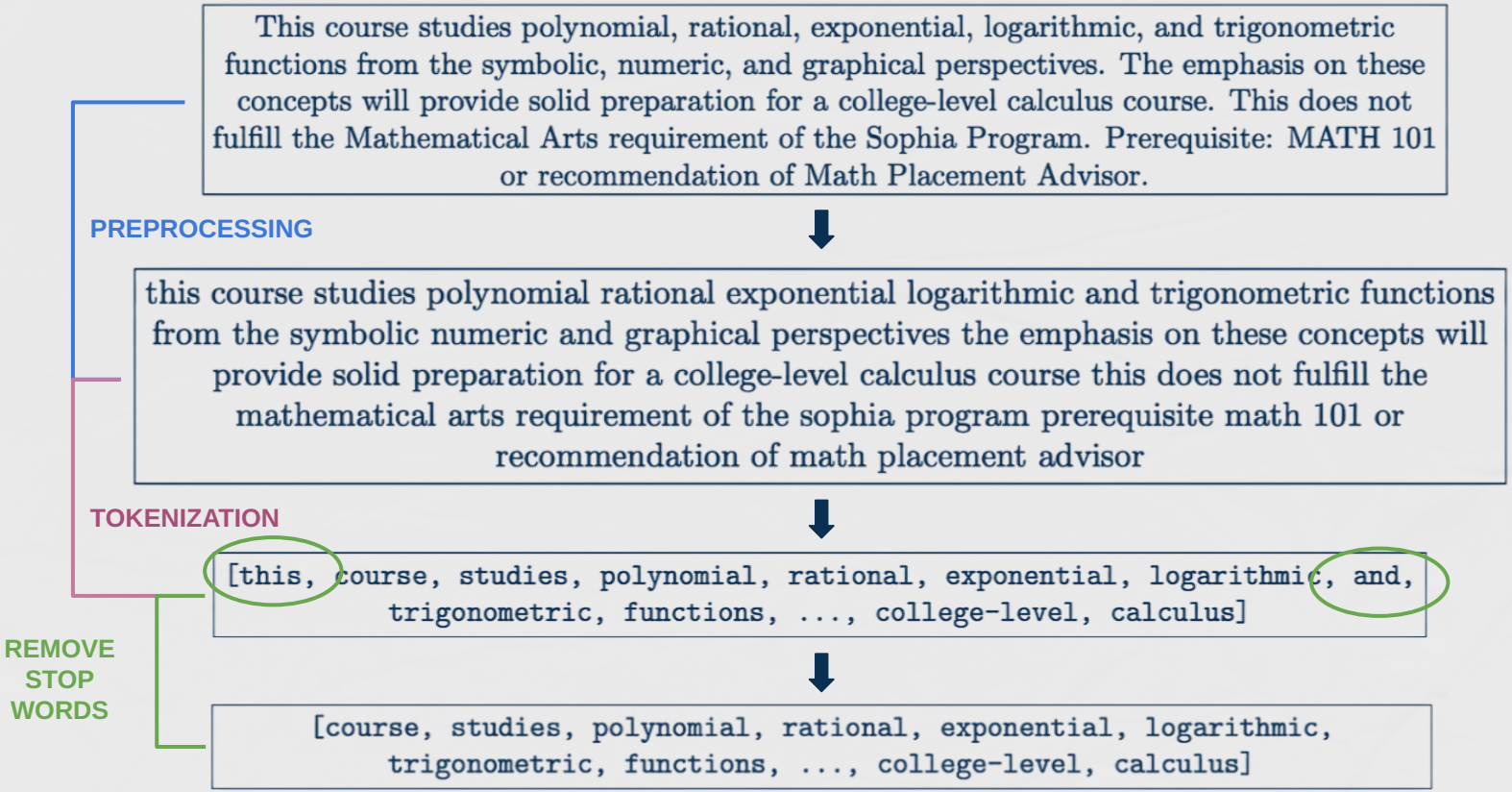
Tokenization

Cleaned text is split into individual words (***tokens***)

Stop Words

After tokenization

Removing very common words


This course studies polynomial, rational, exponential, logarithmic, and trigonometric functions from the symbolic, numeric, and graphical perspectives. The emphasis on these concepts will provide solid preparation for a college-level calculus course. This does not fulfill the Mathematical Arts requirement of the Sophia Program. Prerequisite: MATH 101 or recommendation of Math Placement Advisor.
PREPROCESSING
this course studies polynomial rational exponential logarithmic and trigonometric functions from the symbolic numeric and graphical perspectives the emphasis on these concepts will provide solid preparation for a college-level calculus course this does not fulfill the mathematical arts requirement of the sophia program prerequisite math 101 or recommendation of math placement advisor
TOKENIZATION
[this, course, studies, polynomial, rational, exponential, logarithmic, and, trigonometric, functions, ..., college-level, calculus]
REMOVE STOP WORDS
[course, studies, polynomial, rational, exponential, logarithmic, trigonometric, functions, ..., college-level, calculus]

Vocabulary & Vector Space

Vocabulary

Set of all *unique* tokens that appear across entire training data corpus

Vectorization

Transform the tokens into structured numerical format

1. Count how many times each word appears in a description

2. Arrange the counts into a vector

Bag-of-Words

Each description

- A *collection* of words
- Raw word counts

Ignores word order, grammar, and sentence structure

Example

V = [calculus, polynomial, trigonometric, literature, justice, poetry]

d_1 (MATH 103): This course studies polynomial, rational, exponential, logarithmic, and trigonometric functions from the symbolic, numeric, and graphical perspectives. The emphasis on these concepts will provide solid preparation for a college-level calculus course. This does not fulfill the Mathematical Arts requirement of the Sophia Program. Prerequisite: MATH 101 or recommendation of Math Placement Advisor.

d_2 (ENLT 216): What role do literary works play in advocating for a more just society? Can literature provide a platform for social exploration? An opportunity to engage the emotions of the public? A space for disruption and critique? What are the limitations of literary representations of social justice issues? This course uses these guiding questions to examine works of poetry, fiction, and nonfiction.

Example

bag of words (d1) = [1, 1, 1, 0, 0, 0]
bag of words (d2) = [0, 0, 0, 1, 1, 1]

d_1 (MATH 103): This course studies polynomial, rational, exponential, logarithmic, and trigonometric functions from the symbolic, numeric, and graphical perspectives. The emphasis on these concepts will provide solid preparation for a college-level calculus course. This does not fulfill the Mathematical Arts requirement of the Sophia Program. Prerequisite: MATH 101 or recommendation of Math Placement Advisor.

d_2 (ENLT 216): What role do literary works play in advocating for a more just society? Can literature provide a platform for social exploration? An opportunity to engage the emotions of the public? A space for disruption and critique? What are the limitations of literary representations of social justice issues? This course uses these guiding questions to examine works of poetry, fiction, and nonfiction.

Term Frequency (TF)

Measures how often a word appears within a single document

For a word t and a document d, the term frequency is denoted as

$$tf_{t,d}$$

number of times token *t* appears in document *d*

Term Frequency (TF)

$$tf_{t,d} = \frac{\text{number of times } t \text{ appears in } d}{\text{total number of terms in } d}.$$

Term Frequency (TF)

MATH 103: PRE-CALCULUS

```
[course, studies, polynomial, rational, exponential, logarithmic,
    trigonometric, functions, ..., college-level, calculus]
```

Inverse Document Frequency (IDF)

Measures how common or rare a word is across the entire collection of documents (descriptions)

N = be the total number of documents

df_t = number of documents containing term t

$$idf_t = \log\left(\frac{N}{df_t}\right).$$

TF-IDF Combination

$$\text{tf-idf}_{t,d} = tf_{t,d} \times idf_t$$

TF-IDF as weights

Act as the numeric predictors for the model (each token's weight)

$$x_i = tf_{t,d} \times idf_t$$

TF-IDF Vector

$$|V| = \text{the size of the vocabulary}$$
$$w_{i,d} = \text{TF-IDF weight of term } i \text{ in document } d$$

$$\text{vec}(d) = [w_{1,d}, w_{2,d}, \ldots, w_{|V|,d}],$$

Methods

Preprocessing Pipeline

→ Preprocessing
→ Tokenization
→ Vectorization
→ TF-IDF

Result: TF-IDF vector

→ Logistic Regression

Logistic Regression Model

Probability that a course description belongs to the **Math** category:

Probability P(Y = 1)

Probability that a course description belongs to the **English Literature** category:

Probability P(Y = 0)

Logistic Regression Model

$$p(X) = P(Y = 1 \mid X)$$

where X is the vector of features derived from the text

Objective

Model how these predictors relate to the probability, while ensuring that the output remains within the valid range:

$0 < p < 1$

Logistic Regression Model

Linear Regression

$$Y = \beta_0 + \beta_1 X$$

Taking any value in (-∞, ∞)

Models a continuous outcome variable as a weighted sum of predictors, using a weighted combination of input features

Logistic Regression Model

Natural Logarithm of the odds

$$\text{odds} = \frac{P(Y=1)}{1-P(Y=1)} = \frac{\text{probability of success}}{\text{probability of failure}}$$

Apply natural logarithm to odds → produces the *log-odds* (*logit*)

$$\text{logit}(P(Y=1)) = \ln\left(\frac{P(Y=1)}{1-P(Y=1)}\right)$$

Maps probabilities in (0, 1) to the entire real line (-∞, ∞)

Logistic Regression Model

$$p(X) = \frac{e^{\beta_0+\beta_1 x_1+\cdots+\beta_n x_p}}{1 + e^{\beta_0+\beta_1 x_1+\cdots+\beta_n x_p}}$$

Logistic Regression Model

Coefficients

Each β_i (beta) tells the moel how a word (its TF-IDF weight) pushes the prediction

positive β → more Math-like
negative β → less Math-like

Logistic Regression Model

Odds Ratio

$$\log(\text{odds}) = \beta_0 + \beta_1 x_1 + \ldots$$

Exponentiate

$$\text{odds} = e^{\beta_0 + \beta_a x_1 + \ldots}$$

The quantity e^{β_i} is the *odds ratio* associated with a one-unit increase in x_i:

$$e^{\beta_i} = \frac{\text{odds after a one-unit increase in } x_i}{\text{original odds}}$$

$\beta = 0.7 \rightarrow e^{0.7} \approx 2.01$

Logistic Regression Model

Odds Ratio → Logistic Regression

$$p(X) = \frac{\text{odds}}{1 + \text{odds}}$$



$$p(X) = \frac{e^{\beta_0 + \beta_1 x_1 + \ldots}}{1 + e^{\beta_0 + \beta_1 x_1 + \ldots}}$$

How TF-IDF and Logistic Regression Work Together

TF-IDF	Logistic Regression	Final “influence”
Gives every word a fixed importance score based on how often it appears in the entire collection of documents	Learns separate coefficients (the β’s) that show how strongly each TF-IDF of a word pushes towards Math or away from Math	TF-IDF x β

Training the Model

How the Model Learns

- Estimate the coefficients β_i using *maximum likelihood estimation*
  - finds the set of β values that maximize the probability of concluding the correct labels in the training data
    [Further analysis is beyond scope of project]

Training the Model

Implementation

- ➔ Implement in Python
- ➔ Load dataset from a CSV file
- ➔ Standardize course descriptions
- ➔ Initialize TF-IDF vector
- ➔ Apply vec.fit_transform to learn vocab and convert to vector
- ➔ Map labels (MATH and ENLT) to numeric values + train Logistic Regression model
- ➔ Pair model's coefficients with vectorizer's vocabulary
  - ◆ This gives us the most influential terms for predicting a class

Results

Most Informative Words

Beta coefficients

Top MATH terms:	
equations	(0.194)
rational	(0.202)
prerequisite	(0.210)
topics	(0.215)
emphasis	(0.218)
geometry	(0.219)
modeling	(0.219)
mathematical	(0.274)
linear	(0.288)
math	(0.544)

More likely to be math

Top English terms:	
literature	(-0.325)
literary	(-0.324)
writing	(-0.223)
women	(-0.218)
readings	(-0.205)
contemporary	(-0.200)
animals	(-0.186)
health	(-0.186)
1960s	(-0.181)
writers	(-0.181)

Accuracy

The model predicted every course correctly:

MATH = 1
ENLT = 0

```
## check to see if the courses were correctly classified
# get count of how many were correctly classified and misclassfied
# then the true accruacy value
preds = model.predict(X)
preds
```

```
array([0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1])
```

```
y
```

```
array([0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1])
```

Correct predictions: 20
Incorrect predictions: 0

$$\text{Accuracy} = \frac{\text{correct}}{\text{total}} = 1.0$$

Conceptual Walk Through

Python code in Google Colab

New ENLT Summary

- Text was transformed into a TF-IDF vector
- Only 13 word's from the new ENLT description were in model vocabulary (451 tokens)

- **Prediction**: ENLT
- **Probabilities**: ENLT = 0.548, MATH = 0.542
  - Comes from logistic function

- Top ENLT-leaning word: time (-0.007)
- Top MATH-leaning word: prerequisite (0.032)

- Most contributions were close to 0 → not any big push towards predicting it as MATH rather than ENLT

Sources

Tutorials & Texts

- Basic Text Classification – TensorFlow Core Tutorial
  tensorflow.org/tutorials/keras/text_classification
- Deep Learning with Python (Third Edition) – François Chollet (Manning, 2023)
- Introduction to Information Retrieval (Stanford IR Book) – Manning, Raghavan, Schütze
  nlp.stanford.edu/IR-book
- TF–IDF (Section 6.2), Introduction to Information Retrieval
  nlp.stanford.edu/IR-book/pdf/06vect.pdf

Core Topics

- Text Classification – IBM Think Blog
- Logistic Regression (Baseline Model) – IBM Think Blog
- Train/Validation/Test Splits – V7 Labs
- Importing CSVs into Colab – Stack Overflow; Saturn Cloud

Project Tools & Implementation References

- Google Colab: Importing Files to Notebook – Stack Overflow (2017)
- How to Import Files from Google Drive to Colab – Saturn Cloud (2023)

Thank you!



Q&A TOPICS TO KNOW

Why use log?

1. Balances the influence of rare words
2. Converts multiplicative differences in document frequency into additive differences in log space

IDF as log-odds

$$U_t = \frac{df_t}{N}, \qquad 1 - U_t = \frac{N - df_t}{N}$$

Overall relationship:

$$\text{weights:} \quad \log\left(\frac{1 - U_t}{U_t}\right) \approx \log\left(\frac{N}{df_t}\right) \leftarrow (idf_t)$$

Sparsity

`605 stored elements` means 605 non-zero entries in matrix

Thus, the "sparsity" is then 605/(20*451) = 0.067

Questions

What if a word in the new description is not in the trained model, what will it do?

- It gets no TF-IDF value
- Doesn’t appear in vector
- Has no impact on prediction

What to do?

- Would have to retrain the model with the new description included
- The vocabulary is fixed after training, so we can’t add new words without retraining it

Questions

Documentation for Logistic Regression in scikit-learn
Link

..

Take this slide out or hand draw it

$$DTM = \begin{array}{c|cccccccc} & w_1 & w_2 & w_3 & w_4 & w_5 & w_6 & \cdots & w \\ \hline d_{1t1} & 4 & 0 & 0 & 0 & 1 & 2 & & 0 \\ d_{1t2} & 0 & 1 & 7 & 0 & 0 & 1 & & 3 \\ d_{2t1} & 0 & 5 & 0 & 3 & 0 & 9 & & 1 \\ d_{2t2} & 3 & 0 & 8 & 0 & 1 & 0 & & 0 \\ d_{Jt1} & 0 & 2 & 1 & 0 & 2 & 12 & & 4 \\ d_{Jt12} & 1 & 0 & 0 & 4 & 0 & 0 & & 2 \end{array}$$